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Improving short-term HVAC load forecasting via temporal feature engineering: insights from the PLEIAData building dataset

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Abstract

Building energy consumption accounts for a significant portion of global energy use, with heating, ventilation, and air conditioning (HVAC) systems representing one of the largest contributors to building energy demand. Accurate short-term HVAC load forecasting is therefore essential for efficient building operation, demand-response strategies, and intelligent energy-management systems.This study investigates the impact of temporal feature representation on one-hour-ahead HVAC load forecasting using real operational data from the PLEIAData building dataset. Three forecasting models representing both machine learning and deep learning paradigms—Extreme Gradient Boosting (XGBoost), Long Short-Term Memory (LSTM), and Gated Recurrent Unit (GRU)—are evaluated under progressively enriched temporal feature configurations. The experiments systematically compare baseline environmental variables, autoregressive load features, rolling statistical representations, and extended temporal feature sets designed to capture recent HVAC operational dynamics.The results demonstrate that models relying only on contemporaneous environmental variables exhibit limited predictive capability, whereas incorporating temporal load-history information substantially improves forecasting accuracy. Extended temporal representations based on autoregressive and rolling statistical features consistently enhance model performance across all forecasting architectures. Among the evaluated configurations, XGBoost combined with the extended temporal feature representation achieves one of the best and most stable forecasting performances with an R^2 value of approximately 0.799. Additional ablation and interpretability analyses further show that recent temporal load information is the primary contributor to forecasting performance, while extending the feature space with increasingly complex rolling statistics and larger temporal representations provides only marginal additional gains. Overall, the study highlights the importance of temporal feature design, robustness-oriented evaluation, and interpretable forecasting analysis in HVAC load prediction applications.

Keywords HVAC load forecasting, Temporal feature engineering, Building energy prediction, Autoregressive features, Rolling statistical features, Short-term forecasting, Interpretable forecasting

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Introduction

Buildings represent one of the largest energy-consuming sectors worldwide, accounting for a significant share of global electricity demand. According to the International Energy Agency (IEA), buildings are responsible for nearly one-third of global energy consumption, while electricity demand associated with cooling and air-conditioning continues to increase rapidly in many regions of the world [1, 2]. This trend is expected to accelerate further due to urbanization, population growth, and increasing adoption of cooling technologies, particularly in developing countries [2].

As a result, heating, ventilation, and air-conditioning (HVAC) systems constitute a major component of building energy consumption and represent a critical target for improving energy efficiency and mitigating peak electricity demand [3]. Consequently, enhancing HVAC operational efficiency has become an important research topic within sustainable energy management and smart-building technologies, with increasing attention devoted to predictive HVAC control, intelligent building energy-management systems, and data-driven optimization approaches [4, 5].

Within this context, accurate forecasting of HVAC energy demand plays a key role in enabling advanced control strategies, demand-response applications, and intelligent energy-management systems [6, 7]. In particular, short-term HVAC load forecasting is essential for real-time building operation because it allows energy-management systems to anticipate load variations and perform proactive control actions [8–11]. Recent advances in machine learning and deep learning have significantly improved forecasting capabilities in building-energy applications, especially for nonlinear and time-dependent operational behavior.

However, despite the growing success of data-driven forecasting models, forecasting performance in HVAC systems often depends strongly on how temporal information is represented within the input data. Due to the thermal inertia of buildings and delayed HVAC control responses, recent operational history may contain valuable predictive information for short-term forecasting tasks [12–14]. Consequently, temporal feature representations based on historical load behavior may substantially improve forecasting performance.

Motivated by these considerations, this study investigates the impact of temporal feature representation on one-hour-ahead HVAC load forecasting using real operational data from the PLEIAData building dataset [15]. Three forecasting models representing both machine learning and deep learning paradigms—Extreme Gradient Boosting (XGBoost), Long Short-Term Memory (LSTM), and Gated Recurrent Unit (GRU)—are evaluated under progressively enriched temporal feature configurations. The experiments systematically compare baseline environmental variables, autoregressive load features, rolling statistical representations, and extended temporal feature sets designed to capture recent HVAC operational dynamics.

The results demonstrate that incorporating temporal load-history information substantially improves forecasting performance across all evaluated models. Additional ablation and interpretability analyses further show that recent temporal information remains the dominant contributor to forecasting accuracy, while increasingly complex temporal representations provide limited additional benefit beyond short-term variability descriptors.

The main contributions of this study can be summarized as follows:

1. Systematic investigation of temporal feature representations, including baseline environmental variables, autoregressive load inputs, rolling statistical features, and extended temporal descriptors for one-hour-ahead HVAC load forecasting.
2. Controlled comparative evaluation of machine learning and deep learning models (XGBoost, LSTM, and GRU) under identical temporal input configurations, enabling isolation of temporal representation effects from model-architecture differences.
3. Robustness-oriented ablation analysis examining lag depth, rolling-window size, rolling statistics, and operational proxy variables within HVAC forecasting.
4. Interpretability and diagnostic analysis using permutation feature importance and residual evaluation to examine the relative contributions of temporal and environmental variables to forecasting performance.
5. Empirical evidence demonstrating that carefully engineered temporal representations can substantially influence HVAC forecasting accuracy even under relatively controlled model architectures.

Related work

Machine learning approaches for load forecasting

Machine learning (ML) techniques have been widely applied to building energy forecasting due to their ability to model nonlinear relationships between environmental variables, building operational parameters, and energy demand [16–18]. With the increasing deployment of building automation systems and Internet-of-Things (IoT) sensors, large volumes of operational data have become available, enabling the development of data-driven predictive models for building energy consumption [16, 17].

Traditional machine learning algorithms such as Random Forest (RF), Support Vector Regression (SVR), and gradient boosting methods have been extensively used for building load forecasting tasks [19–23]. These models are capable of capturing nonlinear relationships between weather conditions, occupancy-related variables, and energy consumption patterns. Several studies have demonstrated the effectiveness of ML models in predicting building energy demand under different operational scenarios [20–23]. Recent studies have further explored advanced machine learning frameworks and feature engineering strategies to improve forecasting accuracy in building energy systems [24–27]. Among ensemble learning methods, the Extreme Gradient Boosting (XGBoost) algorithm has gained particular attention due to its strong predictive capability and ability to model complex nonlinear interactions among input variables [27, 28]. These characteristics make XGBoost a competitive approach for short-term load forecasting problems in smart building environments.

Despite their advantages, traditional ML models often require careful feature design and preprocessing in order to achieve reliable prediction performance. The selection and construction of informative input variables therefore play a crucial role in determining the effectiveness of ML-based forecasting models.

Deep learning approaches for load forecasting

In recent years, deep learning (DL) approaches have attracted increasing attention in building energy forecasting research due to their ability to automatically learn complex nonlinear feature representations from large datasets [29]. Deep neural networks are particularly well suited for modeling nonlinear and time-dependent patterns in

building energy consumption, and several studies have demonstrated their effectiveness in improving forecasting accuracy for building energy applications [30–32].

Recurrent neural network (RNN) architectures such as Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks are especially suitable for modeling time-series energy consumption data because they are capable of capturing temporal dependencies in sequential observations [33, 34]. These models have been successfully applied to various building load forecasting tasks, including short-term electricity demand prediction and HVAC energy consumption modeling [35, 36]. Several studies have demonstrated that DL-based forecasting frameworks can effectively capture dynamic load behavior and achieve competitive forecasting performance when sufficient historical data are available [37–40].

Nevertheless, although DL models can automatically extract hierarchical feature representations, their performance can still depend significantly on the quality and relevance of the input variables. As a result, the design of informative temporal features and appropriate input representations remains an important factor even in DL-based forecasting frameworks.

Feature engineering in load forecasting

Feature engineering plays an important role in time-series forecasting problems, particularly in building energy prediction where temporal dynamics strongly influence system behavior. In HVAC systems, recent operational history may contain valuable predictive information due to the thermal inertia of buildings and delayed HVAC control responses [12]. Consequently, temporal representations based on historical load behavior can substantially influence short-term forecasting performance.

In the literature, some studies have used temporal feature engineering approaches such as autoregressive variables, lag-based descriptors, or rolling statistical representations in order to better capture short-term consumption dynamics [13, 14]. These approaches aim to incorporate recent historical information into predictive models and improve the representation of temporal dependencies in building energy consumption patterns.

Recent forecasting studies further suggest that forecasting improvements may arise not only from increasingly sophisticated model architectures, but also from enhanced temporal input representations and engineered forecasting descriptors [24–27]. For example, Incremona et al. [24] demonstrated that enhanced predictive representations and nonlinear expert aggregation strategies can improve electricity load forecasting performance even without substantially increasing model complexity. Similarly, Shafiuzzaman et al. [26] showed that multi-lag feature engineering can significantly improve very short-term load forecasting accuracy by better capturing recent temporal dependencies and operational dynamics. Recent reviews on HVAC energy forecasting further emphasize that lagged variables, temporal indicators, weather-related descriptors, and contextual input representations remain among the most widely adopted and effective forecasting inputs in practical building energy prediction systems [25]. In addition, comparative forecasting studies combining XGBoost and deep learning models indicate that forecasting robustness often depends strongly on temporal input design and feature preprocessing strategies rather than solely on model architecture complexity [27].

These observations suggest that temporal representation design remains a critical component of short-term load forecasting frameworks. Nevertheless, relatively few

studies systematically investigate how progressively enriched temporal representations influence forecasting behavior across both machine learning and deep learning architectures under controlled experimental conditions. This study addresses this gap by comparatively evaluating autoregressive and rolling statistical temporal representations across multiple forecasting architectures using identical input configurations.

Dataset description

The dataset used in this study is the PLEIAData building energy dataset, which contains operational measurements collected from the Pleiades building complex located at the University of Murcia, Spain. The dataset was introduced by Ibarra et al. and provides detailed information on building operation, HVAC system activity, and environmental conditions measured in a real smart-building environment [15]. Table 1 summarizes the variables available in the PLEIAData dataset.

The original dataset includes both raw measurements and processed versions of the dataset. In this study, the processed version provided by the dataset authors was used, containing cleaned and synchronized measurements suitable for forecasting analysis. The Pleiades complex consists of three adjacent buildings (A, B, and C). Since these buildings exhibit similar construction characteristics and operational behavior, the experiments were conducted using only Building A in order to avoid redundant modeling while preserving representative HVAC operational characteristics.

PLEIAData provides measurements at multiple temporal resolutions, including both 10-minute and hourly (60-minute) aggregated data. In this study, the hourly-resolution dataset corresponding to Building A was selected. The choice of hourly resolution is motivated by two main considerations. First, one-hour-ahead forecasting is widely adopted in practical HVAC scheduling, building energy management, and demand-response applications. Second, hourly aggregation reduces high-frequency fluctuations and facilitates clearer analysis of short-term operational dynamics and temporal feature representations. Therefore, the forecasting problem considered in this study is defined as one-hour-ahead HVAC load prediction, which remains consistent with common practice in short-term building energy forecasting literature [7–11, 25].

The dataset covers the period between January 1 and December 18, 2021, representing nearly a full year of building operation. It integrates multiple sources of information, including indoor environmental conditions, outdoor weather variables, HVAC operational states, and building energy consumption. These measurements enable comprehensive analysis of building energy behavior and provide a valuable resource for data-driven research in smart building energy systems.

PLEIAData contains a diverse set of environmental and operational variables describing building conditions and HVAC system behavior. These include indoor temperature ($V2$), outdoor temperature $(tmed)$, solar radiation $(radmed)$, HVAC operational indicators such as the number of active devices $(V4)$, as well as control-related variables (e.g., $V12$) and additional meteorological indicators such as humidity, wind speed, precipitation, and derived variables such as dew point temperature and vapor pressure deficit. The forecasting target in this study is the hourly HVAC energy consumption, denoted as dif_cons_real, which represents the energy demand associated with HVAC system operation. All variables listed in Table 1 are treated as candidate predictors, while the

Table 1 Main variables available in the PLEIAData building energy dataset

Variable	Category (Unit)	Variable type	Description
dif_cons_real	HVAC load (kWh)	Target variable	Real hourly HVAC energy consumption used as the prediction target
$cons_total$	Energy consumption (kWh)	Candidate variable	Cumulative consumption variable with limited suitability for short-term forecasting tasks
dif_cons_smooth	HVAC load (kWh)	Candidate variable	Smoothed version of the load signal
$V2$	Indoor environment (°C)	Candidate variable	Indoor temperature influencing HVAC demand
$V4$	HVAC operation (count)	Candidate variable	Number of active HVAC devices reflecting system operation
$V12$	HVAC control (°C)	Candidate variable	HVAC setpoint-related control parameter
$V26$	HVAC system type (categorical)	Candidate variable	Categorical descriptor of HVAC system configuration
$V5_1$	HVAC mode (binary)	Candidate variable	Binary heating mode indicator
$V5_2$	HVAC mode (binary)	Candidate variable	Binary cooling mode indicator
Hour_1–Hour_3	Time indicators (categorical)	Candidate variable	Encoded time-of-day indicators
Season_1–Season_4	Seasonal indicators (categorical)	Candidate variable	Encoded seasonal information
$tmed$	Outdoor environment (°C)	Candidate variable	Outdoor temperature affecting HVAC load
$hrmed$	Weather (%)	Candidate variable	Relative humidity indicator
$radmed$	Weather (W/m^2)	Candidate variable	Solar radiation contributing to heat gains
$vvmed$	Weather (m/s)	Candidate variable	Wind speed
$dvmed$	Weather (° from north)	Candidate variable	Wind direction
$prec$	Weather (l/m^2)	Candidate variable	Precipitation
$dewpt$	Weather (°C)	Candidate variable	Dew point temperature
dpv	Weather (kPa)	Candidate variable	Vapor pressure deficit indicator

final feature selection is performed in Sect. "Exploratory Data Analysis (EDA)" based on statistical analyses and modeling considerations.

Short-term load dynamics

To illustrate the short-term dynamics of HVAC energy consumption, Fig. 1 shows a representative one-week segment of HVAC load together with indoor and outdoor temperature measurements from the PLEIAData dataset. As shown in Fig. 1, HVAC energy demand closely follows variations in outdoor temperature, with higher loads occurring during warmer periods. This behavior reflects the expected response of the cooling system to increased external thermal stress.

Although HVAC load generally follows outdoor temperature trends, some periods exhibit deviations from this pattern. For example, around July 4th, elevated outdoor temperature does not lead to a proportional increase in HVAC energy consumption. Such behavior may reflect operational or occupancy-related variations during weekends or low-activity periods, highlighting the influence of operational factors on HVAC load dynamics. In contrast, the indoor temperature measurement variable $(V2)$ remains relatively stable throughout the selected period despite significant fluctuations in outdoor conditions. Although the absolute temperature levels are relatively high during the summer period, the stability of the signal suggests consistent indoor thermal behavior within the monitored building environment.

At the same time, the HVAC load signal exhibits clear short-term persistence and daily cyclic patterns. These temporal characteristics suggest that recent system states influence near-future energy demand. Therefore, incorporating temporal dependencies into forecasting models becomes essential. In particular, lagged and rolling features can capture the thermal inertia of the building and the operational continuity of the HVAC system, motivating their use in the feature engineering stage.

Exploratory data analysis (EDA)

Descriptive statistics

This section provides descriptive statistics of the candidate variables in the dataset prior to feature selection. The objective is to offer an initial understanding of the data distribution, scale, and variability of key variables. Descriptive statistics were computed for all candidate variables; however, only a representative subset is reported for clarity. These

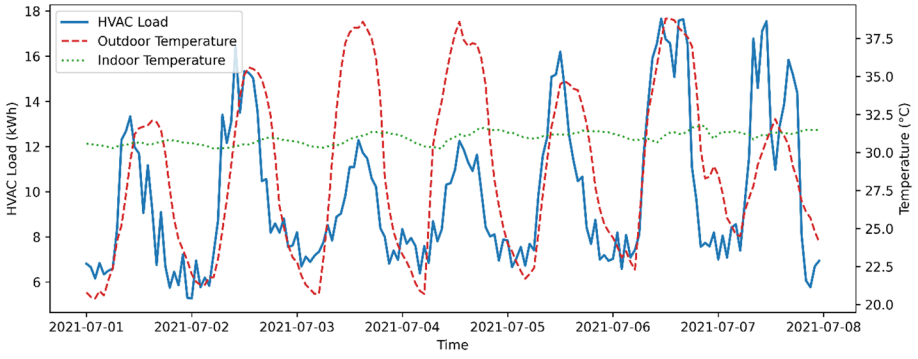


Fig. 1 Short-term dynamics of HVAC energy consumption together with the indoor temperature measurement variable ($V2$) and outdoor temperature ($tmed$) over a representative one-week period from the PLEIAData dataset

variables are presented for illustrative purposes and do not imply final inclusion in the forecasting models. A more comprehensive feature relevance analysis is conducted in the following sections using both linear (correlation) and nonlinear (mutual information) methods.

Table 2 presents the descriptive statistics of a representative subset of variables in the PLEIAData dataset, including HVAC load (dif_cons_real), environmental variables $(tmed, radmed, dewpt)$, operational indicators $(V2, V4)$, control-related variables $(V12)$, system-level characteristics $(V26)$, and seasonal indicators $(Season_3)$.

The results indicate that HVAC load exhibits substantial variability (mean = 5.019 kWh, std = 3.044 kWh), reflecting the dynamic nature of building energy consumption.

Among environmental variables, outdoor temperature $(tmed)$ spans a wide range (0–43.7 °C), capturing seasonal variability, while solar radiation $(radmed)$ shows high dispersion due to daily and weather-related fluctuations. The dew point temperature $(dewpt)$, representing humidity-related conditions, also exhibits significant variability, indicating that moisture-related effects may play a role in HVAC demand.

Indoor and operational variables such as $V2$ and $V4$ demonstrate variability associated with system usage patterns and building operation. The HVAC control variable $V12$ exhibits relatively low variance, suggesting that setpoint control operates within a constrained range; however, such variables may still contribute to prediction through nonlinear interactions.

The variable $V26$ represents the type of HVAC device (wall-mounted or ceiling-mounted), originally defined as a categorical variable. In the processed dataset, it is numerically encoded, resulting in a limited but non-zero variance. Although its variability is relatively small, such system-level characteristics may still contribute to prediction performance through nonlinear interactions with environmental and operational variables. The seasonal indicator ($Season_3$) reflects categorical temporal patterns and highlights the presence of calendar-based variability in the dataset.

Overall, the descriptive statistics confirm that the dataset contains a diverse set of environmental, operational, and temporal variables with varying degrees of variability. While these statistics provide an initial understanding of the data, a more rigorous assessment of feature relevance is performed in the following sections using both linear (correlation) and nonlinear (mutual information) analysis.

Correlation analysis

To ensure a comprehensive evaluation, Pearson correlation analysis was initially conducted on the full set of candidate variables listed in Table 1. For clarity and visualization purposes, Fig. 2 presents a representative subset of variables selected based on their relevance to HVAC load and their roles in capturing environmental conditions, system operation, control signals, system characteristics, and seasonal effects. These variables include indoor temperature $(V2)$, outdoor temperature $(tmed)$, solar radiation $(radmed)$, the number of active HVAC devices $(V4)$, the setpoint-related HVAC control parameter $(V12)$, the system-level variable representing device type $(V26)$, a seasonal indicator $(Season_3)$, and weather-derived variables (dew point temperature $(dewpt)$ and vapor pressure deficit (dpv)), along with the HVAC load target (dif_cons_real).

Table 2 Descriptive statistics of a representative subset of variables in the PLEIAData dataset (hourly resolution)

Variable	Category	Description (Unit)	Mean	Std	Min	Max
$V2$	Indoor	Indoor air temperature (°C)	27.615	3.406	19.177	33.934
$V12$	HVAC control	Setpoint-related control variable (°C)	18.530	1.441	15.630	21.240
$tmed$	Outdoor	Outdoor air temperature (°C)	19.236	7.293	0.000	43.700
$radmed$	Outdoor	Solar radiation (W/m^2)	196.150	279.367	0.000	1078.700
$V4$	HVAC control	No. of active HVAC devices (count)	7.307	2.957	2.000	17.000
$V26$	System	Type of HVAC device (encoded numerically)	0.297	0.120	0.100	0.800
$dewpt$	Weather	Dew point temperature (°C)	9.709	5.773	-8.000	21.800
$Season_3$	Calendar	Seasonal indicator (binary)	0.278	0.448	0.000	1.000
dif_cons_real	Target	HVAC energy consumption (kWh)	5.019	3.044	0.200	24.910

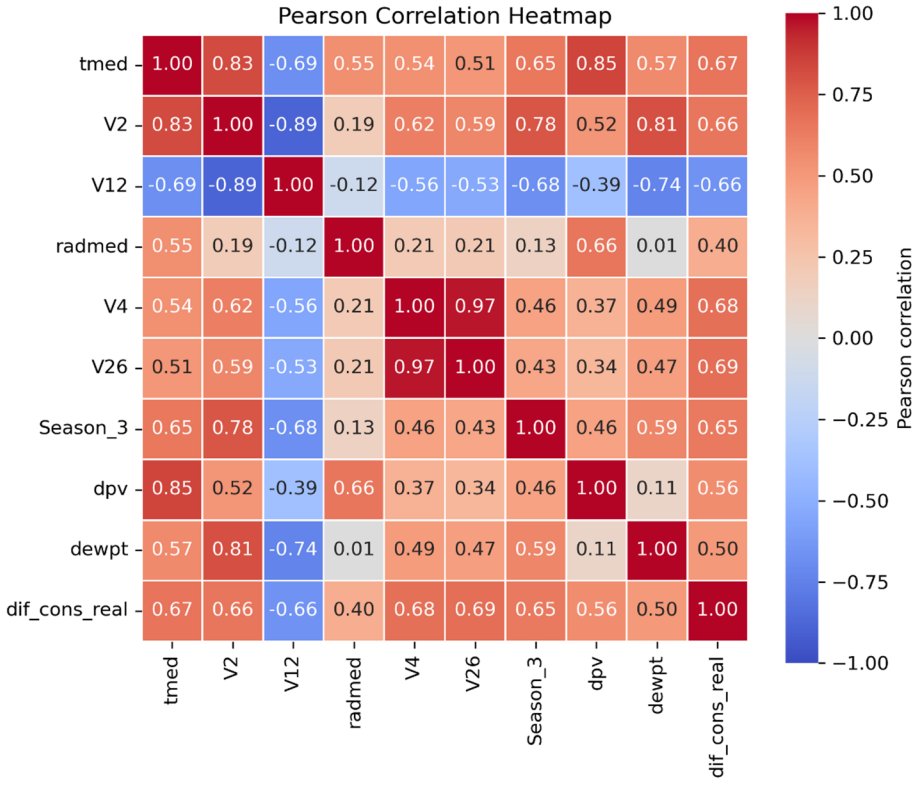


Fig. 2 Pearson correlation heatmap of a representative subset of candidate variables, including environmental, operational, control, system-level, and seasonal features. The figure highlights both the relationships between predictors and the HVAC load target (dif_cons_real) and the strong interdependencies among predictors, particularly the near-perfect correlation between $V4$ and $V26$, indicating redundancy between system configuration and operational activity

The correlation heatmap indicates that multiple variables exhibit moderate to strong linear relationships with HVAC energy consumption, confirming that building energy demand is influenced by a combination of environmental, operational, and control-related factors. In particular, device type $(V26)$, the number of active HVAC devices $(V4)$, outdoor temperature $(tmed)$, indoor temperature $(V2)$, and the seasonal indicator $(Season_3)$ show moderate positive correlations with HVAC load (0.69, 0.68, 0.67, 0.66, and 0.65, respectively). In contrast, the HVAC control variable $(V12)$ exhibits a moderate negative correlation (-0.66), reflecting its regulatory role in system operation.

In addition to their relationships with the target variable, the predictors themselves exhibit strong interdependencies. For example, outdoor temperature $(tmed)$ is strongly correlated with indoor temperature $(V2)$ (0.83) and vapor pressure deficit (dpv) (0.85), while indoor temperature $(V2)$ shows a strong negative correlation with the control variable $(V12)$ (-0.89) and strong positive correlations with both seasonal indicators and dew point temperature $(dewpt)$. These relationships indicate that temperature-related and humidity-related variables capture overlapping physical information associated with the building's thermal environment.

A key observation from Fig. 2 is the near-perfect positive correlation between the number of active HVAC devices $(V4)$ and the system-level variable representing device type $(V26)$. This indicates that system configuration and operational activity are tightly coupled in the dataset. In practical terms, buildings equipped with similar device types tend to exhibit consistent activation patterns, leading to a strong statistical dependency

between system configuration and operational intensity. This suggests that $V26$ may not provide entirely independent information but instead reflects underlying operational patterns already captured by $V4$.

From a modeling perspective, this redundancy implies that including both variables may introduce multicollinearity without substantially improving predictive performance. This observation is consistent with the experimental results, where the inclusion of $V26$ leads to only marginal improvements compared to configurations already containing $V4$. Therefore, feature selection should consider not only the strength of correlation with the target variable but also the dependencies among predictors in order to construct a compact and informative feature set.

However, it is important to emphasize that Pearson correlation captures only linear dependencies between variables. As a result, variables that appear weakly correlated with the target may still contain predictive information through nonlinear relationships or interaction effects. To address this limitation, a complementary nonlinear feature relevance analysis based on mutual information is performed in the following section.

Mutual information

To complement the linear correlation analysis, a nonlinear feature relevance analysis was conducted using mutual information (MI). Unlike Pearson correlation, mutual information captures both linear and nonlinear dependencies between variables, making it suitable for identifying complex relationships and interaction effects that may not be revealed through linear analysis.

In this study, mutual information scores were computed between candidate variables at time t and the HVAC load at the next time step $y_{(t+1)} = dif_cons_real_{(t+1)}$, allowing the evaluation of predictive relevance in a one-step-ahead forecasting framework. The results, presented in Fig. 3, show that the current HVAC load $y_{(t)} = dif_cons_real_{(t)}$, exhibits the highest mutual information with the target. This is expected, as the HVAC system exhibits strong temporal persistence due to building thermal inertia and operational continuity. Nevertheless, this result quantitatively confirms that short-term HVAC load dynamics are primarily governed by recent system states.

Importantly, beyond the autoregressive component, several additional variables demonstrate significant nonlinear relevance. In particular, the HVAC control parameter $(V12)$ and indoor temperature $(V2)$ show strong mutual information scores, indicating that control signals and internal thermal conditions play a critical role in shaping load evolution. Furthermore, the number of active HVAC devices $(V4)$, the system-level variable $(V26)$, and outdoor temperature $(tmed)$ also exhibit meaningful contributions.

These findings provide an important extension to the correlation analysis: while some variables (e.g., $V12$ and $V26$) appeared less prominent or ambiguous under linear correlation, they exhibit higher relevance under mutual information. This suggests the presence of nonlinear dependencies and interaction effects, particularly between control signals, system characteristics, and environmental conditions. In contrast, seasonal indicators and weather-derived variables, including dew point temperature $(dewpt)$, vapor pressure deficit (dpv), and solar radiation $(radmed)$, exhibit lower but non-negligible mutual information scores. These variables are therefore interpreted as providing complementary contextual information rather than acting as primary drivers of short-term load variability.

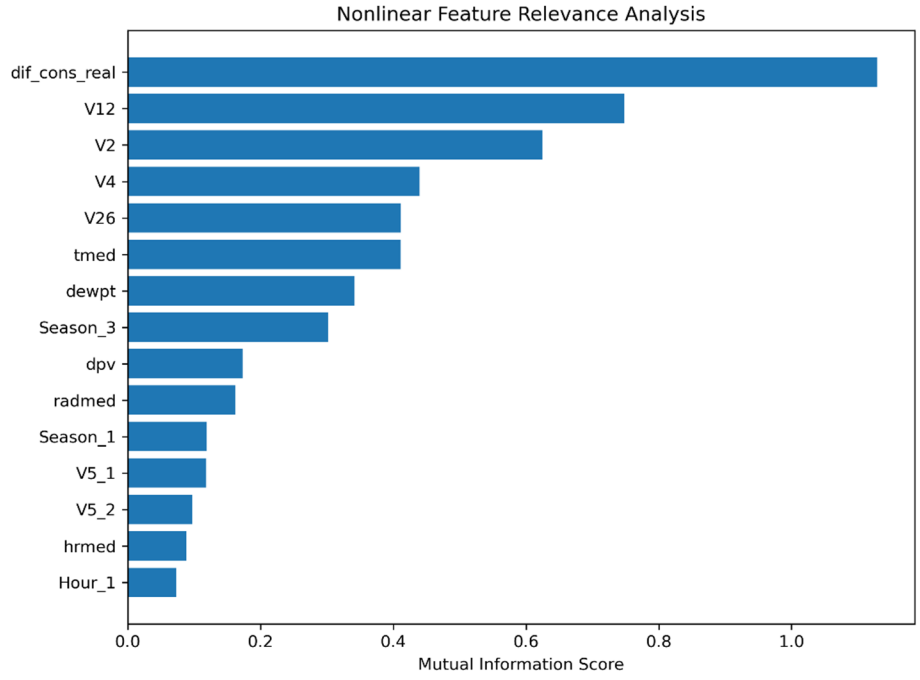


Fig. 3 Nonlinear feature relevance analysis of raw candidate variables using mutual information. The scores were computed between candidate variables at time t and the HVAC load at the $t+1$

Overall, the mutual information analysis demonstrates that HVAC load prediction is influenced by a combination of autoregressive, environmental, operational, and control-related variables. However, even when nonlinear dependencies are taken into account, the predictive structure remains strongly dominated by recent load history. This finding supports the central premise of this study: temporal dependencies encoded through recent load observations play a more significant role than individual exogenous variables in short-term HVAC load forecasting.

Feature selection and engineering

Baseline feature sets

Based on the insights obtained from both correlation and mutual information analyses, a structured and hierarchical feature selection strategy was adopted to construct baseline feature sets with varying levels of complexity. The correlation analysis revealed several groups of highly correlated variables representing similar physical phenomena. Notably, strong pairwise relationships were observed between indoor temperature and HVAC control signals $(V2 - V12)$, operational and system-level variables $(V4 - V26)$, and temperature-related weather indicators (e.g., $tmed - dpv$ and $V2 - dewpt$). These relationships indicate that multiple variables capture overlapping information about the building's thermal state, system operation, and environmental conditions.

To account for this redundancy while preserving physical interpretability, three nested baseline feature sets were defined using different selection criteria as summarized in Table 3.

In the compact feature set (Core-S), only one representative variable was selected from each highly correlated group, prioritizing variables with clear physical interpretation and consistent relevance across both correlation and mutual information analyses.

Table 3 Baseline feature configurations used in the HVAC load forecasting experiments. All baseline variables are defined for the current time t in the model

Core Set	Description	Variables	Number of Features
Core-S (Compact)	Minimal set with one representative from each highly correlated group	$tmed$, $V4$, $V12$, $radmed$	4
Core-M (Balanced)	Moderately expanded set including additional environmental and seasonal variables	$tmed$, $V4$, $V12$, $V2$, $radmed$, $dewpt$, $Season_3$	7
Core-L (Extended)	Full set including variables selected based on mutual information relevance	$tmed$, $V4$, $V12$, $V2$, $V26$, $radmed$, $dewpt$, $Season_3$, dpv	9

This results in a minimal and interpretable feature set capturing the dominant thermal, operational, and control dynamics. In the balanced feature set (Core-M), only near-perfect correlations were treated as exclusion criteria. Moderately correlated variables were retained when they represent distinct physical mechanisms, such as environmental conditions, control signals, and seasonal effects. This set aims to capture additional contextual information without introducing excessive redundancy. In the extended feature set (Core-L), correlation-based constraints were relaxed, and variables identified as relevant through mutual information analysis were included. This configuration represents an upper-bound scenario in terms of feature richness, allowing the evaluation of whether additional variables provide meaningful predictive improvements. This hierarchical design enables a systematic evaluation of the trade-off between model complexity and predictive performance, while explicitly accounting for redundancy and nonlinear feature relevance. In the model baseline variables are defined for the current time t.

Autoregressive features

Autoregressive (AR) features are introduced to capture the short-term persistence commonly observed in HVAC energy consumption. Let $y_{(t)}$ denote the HVAC load at time step *t*. The next time-step load $y_{(t+1)}$ is modeled as

$$y_{(t+1)} = f(x_{(t)}, y_{(t)})$$

where $x_{(t)}$ represents the environmental and operational variables measured at time t. The autoregressive feature was added to baseline feature configurations:

(i) Core-S + AR: $\{tmed,\ V4,\ V12,\ radmed,\ dif_cons_real\}$,
(ii) Core-M + AR: $\{tmed,\ V4,\ V12,\ V2,\ radmed,\ dewpt,\ Season_3,\ dif_cons_real\}$,
(iii) Core-L + AR: $\{tmed,\ V4,\ V12,\ V2,\ V26,\ radmed, dewpt,\ Season_3,\ dpv,\ dif_cons_real\}$

Including the current load $y_{(t)}$ as an input variable allows the models to capture the strong temporal continuity of HVAC demand as well as the delayed thermal response of the building. This reflects the inherent thermal inertia of building systems, where past energy consumption influences future load dynamics. In the feature tables, this variable is denoted as *dif_cons_real*, which represents the most recent observed HVAC load available to the forecasting model.

Rolling statistical features

In addition to autoregressive features, rolling statistical features were introduced to capture short-term temporal trends and smooth fluctuations in both the HVAC load and the input variables.

Instead of considering only the instantaneous values of the variables, a rolling mean representation was computed using the most recent three observations. For a generic variable $z\,(t)$, the rolling mean is

$$roll3_z_{(t)} = \frac{1}{3}\left(z_{(t)} + z_{(t-1)} + z_{(t-2)}\right)$$

This formulation was applied not only to the HVAC load but also to all selected environmental, operational, and control variables. As a result, the model receives both instantaneous measurements and their short-term averaged representations.

The inclusion of rolling features enables the models to capture short-term dynamics and temporal patterns more effectively by reducing noise and emphasizing recent trends. In the context of building energy systems, this is particularly important due to the delayed thermal response and gradual evolution of system states. The rolling features were incorporated on top of the autoregressive configurations, resulting in the final temporal feature sets used in the forecasting experiments. For each baseline core set, rolling features were computed for load and all included baseline variables, forming an extended feature representation that combines instantaneous, autoregressive, and smoothed temporal information.

Forecasting models

To evaluate the effectiveness of temporal feature engineering for HVAC load prediction, three forecasting models representing both ML and DL paradigms were considered. XGBoost was employed as a strong machine learning baseline for tabular data modeling, while Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks were used to capture temporal dependencies in the HVAC load time series.

In this study, temporal dependencies are primarily represented through engineered temporal features rather than long input sequences, allowing a direct comparison between tree-based models and recurrent neural networks under identical input configurations.

Long short-term memory (LSTM)

LSTM networks are an advanced type of recurrent neural network (RNN) specifically designed to capture long-term dependencies in sequential data. Conventional recurrent neural networks often encounter the vanishing gradient problem during training, which makes it difficult to learn relationships over long temporal horizons. LSTM networks mitigate this limitation through a gated memory structure composed of input, forget, and output gates that control the flow of information within the network.

Let $\{x_{(1)}, x_{(2)}, \ldots, x_{(T)}\}$ denote an input sequence, where each element $x_{(t)}?\mathbb{R}^k$ represents a *k*-dimensional feature vector at time step t. The hidden state of the LSTM unit at time t is denoted by $h_{(t)}$, while $h_{(t-1)}$ corresponds to the hidden state generated at the previous time step.

A key component of the LSTM is the memory cell state, which stores information across time steps and enables the network to preserve long-term dependencies. The cell state at time $t-1$, denoted by $s_{(t-1)}$, interacts with the current input $x_{(t)}$ and the previous hidden state $h_{(t-1)}$ to determine how information should be updated, retained, or discarded. This mechanism allows the network to selectively preserve useful information while eliminating irrelevant data. The update of the cell state can be expressed as

$$s_{(t)} = f_t \odot s_{(t-1)} + i_{(t)} \odot g_{(t)}$$

where $\odot$ denotes element-wise multiplication. In this formulation, $g_{(t)}$, $i_{(t)}$ and $f_{(t)}$ represent the candidate input node, the input gate, and the forget gate, respectively.

These quantities are computed from the current input $x_{(t)}$ and the previous hidden state $h_{(t-1)}$ as follows:

$$g_{(t)} = \phi\ \left(W_{gx}x_{(t)} + W_{gh}h_{(t-1)} + b_g\right),$$

$$i_{(t)} = \sigma\ \left(W_{ix}x_{(t)} + W_{ih}h_{(t-1)} + b_i\right),$$

$$f_{(t)} = \sigma\ \left(W_{fx}x_{(t)} + W_{fh}h_{(t-1)} + b_f\right),$$

where W_{gx}, W_{gh}, W_{ix}, W_{ih}, W_{fx}, and W_{fh} denote the weight matrices associated with the corresponding inputs, and b_g, b_i, b_f represent bias vectors. The function $\sigma\,(.)$ corresponds to the sigmoid activation function, while $\phi\,(.)$ denotes the hyperbolic tangent $(tanh)$ activation.

The output gate of the LSTM cell is defined as

$$o_{(t)} = \sigma\ \left(W_{ox}x_{(t)} + W_{oh}h_{(t-1)} + b_o\right),$$

Finally, the hidden state at time t, which represents the output of the LSTM unit, is computed as

$$h_{(t)} = o_{(t)} \odot\ \phi\ \left(s_{(t)}\right)$$

The hidden state $h_{(t)}$ is subsequently used either as input to the next time step or as the output representation for downstream prediction layers. These gating mechanisms enable the network to selectively preserve relevant historical information while filtering out less informative signals. Consequently, LSTM architectures are particularly effective for time-series forecasting tasks in which previous observations influence future system behavior.

In building energy forecasting applications, LSTM models are capable of learning nonlinear relationships between environmental variables, operational states, and temporal features derived from historical observations. In this study, temporal information is incorporated through engineered autoregressive and rolling features that are provided as inputs to the LSTM model.

Gated recurrent unit (GRU)

The Gated Recurrent Unit (GRU) is a recurrent neural network architecture designed to model sequential data while reducing the structural complexity of LSTM networks. GRU was introduced as a simplified variant of LSTM that preserves the ability to learn temporal dependencies but requires fewer parameters and computational resources. By combining certain gating operations, GRU eliminates the explicit cell state used in LSTM and instead maintains a single hidden state to represent the network memory.

Similar to other recurrent architectures, the GRU processes sequential input data $\left\{x_{(1)}, x_{(2)}, \ldots, x_{(T)}\right\}$ where $x_{(t)}?\mathbb{R}^k$ denotes the input feature vector at time step t . The hidden state of the GRU unit at time t is represented by $h_{(t)}$, which contains the encoded information from both the current input and previous states.

The GRU architecture introduces two gating mechanisms: the update gate and the reset gate, which regulate the flow of information through the network.

The update gate determines how much of the previous hidden state should be preserved and is defined as

$$z_{(t)} = \sigma\ \left(W_{zx}x_{(t)} + W_{zh}h_{(t-1)} + b_z\right).$$

The reset gate controls the influence of the previous hidden state when computing new candidate information:

$$r_{(t)} = \sigma\ \left(W_{rx}x_{(t)} + W_{rh}h_{(t-1)} + b_r\right).$$

Using the reset gate, a candidate hidden state is calculated as

$$\widetilde{h}_{(t)} = \phi\ \left(W_{hx}x_{(t)} + W_{hh}\left(r_{(t)} \odot\ h_{(t-1)}\right) + b_h\right).$$

Finally, the hidden state of the GRU unit is updated by combining the previous hidden state and the candidate hidden state according to the update gate:

$$h_{(t)} = \left(1 - z_{(t)}\right) \odot\ h_{(t-1)} + z_{(t)} \odot\ \widetilde{h}_{(t)}\ ,$$

where $\sigma\left(.\right)$ represents the sigmoid activation function, $\phi\left(.\right)$ denotes the hyperbolic tangent ($tanh$) activation, and $\odot$ indicates element-wise multiplication.

Compared with LSTM networks, GRU employs a more compact architecture by merging the memory and hidden state representations into a single state variable. This simplification reduces the number of trainable parameters and often results in faster training while maintaining comparable predictive performance.

In the context of building energy forecasting, GRU models can effectively learn temporal patterns related to HVAC operation, environmental variations, and historical load dynamics. In this study, GRU networks are used to model the nonlinear relationships between environmental variables and the engineered temporal features derived from historical load observations.

Extreme gradient boosting (XGBoost)

Extreme Gradient Boosting (XGBoost) is an ensemble learning algorithm based on gradient boosted decision trees and is widely used for regression and classification tasks due to its strong predictive performance and computational efficiency. The method constructs a sequence of decision trees in a stage-wise manner, where each new tree aims to correct the prediction errors produced by the previously trained trees.

For a dataset consisting of n samples and m input features, the predicted output $\widehat{y}_i$ for the ith observation can be expressed as

$$\widehat{y}_i = \sum_{k=1}^{K} f_k\left(x_i\right),$$

where $f_k(\cdot\ )$ represents the kth regression tree and K denotes the total number of trees in the ensemble. Each tree belongs to the functional space of regression trees, where $f_k\left(x\right)$ maps an input vector x_i to a corresponding leaf weight. The training objective of XGBoost is defined as the minimization of a regularized loss function

$$\mathcal{L} = \sum_{i=1}^{n} l(y_i, \widehat{y}_i) + \sum_{k=1}^{K} \Omega\ (f_k)\,,$$

where $l(.)$ denotes the loss function measuring the discrepancy between the predicted and true values, and $\Omega\ (f_k)$ represents the regularization term used to control model complexity and prevent overfitting.

The regularization component is given by

$$\Omega\ (f_k) = \gamma\, T + \frac{1}{2}\lambda\, ?w?^2,$$

where T denotes the number of leaves in the tree, w represents the vector of leaf weights, and γ and λ are regularization parameters.

During the training process, XGBoost applies gradient boosting to iteratively optimize the objective function. At each iteration, a new decision tree is constructed based on the gradients of the loss function with respect to the current predictions, allowing the model to progressively reduce the prediction error.

Due to its ability to capture nonlinear relationships and complex interactions among input variables, XGBoost has been successfully applied in various energy forecasting studies. In the context of this study, the XGBoost model is employed to learn the relationship between HVAC load and the input variables, which include meteorological parameters as well as engineered temporal features derived from historical load observations.

In this study, XGBoost serves as the primary machine learning baseline model. Unlike recurrent neural networks, XGBoost does not explicitly process sequential inputs. Instead, temporal dependencies are represented through engineered features such as autoregressive variables and rolling statistics. This makes XGBoost particularly suitable for evaluating the effectiveness of temporal feature engineering strategies.

Experimental setup

To evaluate the predictive performance of the proposed forecasting framework, a series of controlled experiments were conducted using the PLEIAData dataset. The experiments aim to assess the impact of temporal feature engineering strategies on HVAC load forecasting across different model architectures.

Train–test split and evaluation protocol

Prior to model training, the dataset was chronologically divided into training and testing subsets. Since the data represent a time series, a random split would violate the temporal ordering of the observations and potentially introduce information leakage. Therefore, a time-ordered split was applied, where the first 80% of the observations were used for training and the remaining 20% were reserved for testing.

This strategy ensures that the models are evaluated using future observations that were not available during training, thereby avoiding information leakage between training and testing sets. The forecasting task is formulated as a one-step-ahead prediction problem, where the HVAC load at time $t + 1$ is predicted using the input variables available at time t.

To further strengthen the robustness of the evaluation and reduce dependence on a single train–test boundary, an expanding-window walk-forward validation was additionally implemented. In this approach, the model is trained on progressively increasing historical data and evaluated on the immediately following unseen time segments. Specifically, multiple temporal folds were constructed, where each fold uses all available past observations for training and a subsequent time block for testing. This process preserves the chronological structure of the data while enabling performance evaluation across different temporal segments. The final performance metrics are reported as the mean and standard deviation of RMSE, MAE, and R^2 across all folds, providing a more reliable assessment of model generalization under varying temporal conditions.

Leakage-free feature engineering pipeline

While the temporal feature engineering strategy is described in Sect. "Feature Selection and Engineering", it is equally important to ensure that these features are generated and evaluated within a strictly causal forecasting framework.

In this study, the entire feature-generation and evaluation workflow was implemented to prevent the use of future information at any stage of the pipeline. As illustrated in Fig. 4, and summarized in Algorithm 1, all input features are constructed using observations available up to time t, while the prediction target corresponds to the next time step $y_{(t+1)}$.

After feature construction and target definition, the dataset is chronologically partitioned using either a fixed train–test split or an expanding-window walk-forward validation scheme. This ensures that all test samples correspond to future observations that are not accessible during model training.

Within each split or fold, preprocessing operations are applied in a leakage-free manner. In particular, for models requiring feature scaling (LSTM and GRU), the scaler is fitted exclusively on the training data and subsequently applied to the test data without refitting. As shown in Fig. 4 and Algorithm 1, this procedure guarantees that no information from the test set influences the training phase.

The workflow consists of the following sequential steps:

(1) Construct temporal features using information up to time t,
(2) Define the prediction target as $y_{(t+1)}$),
(3) Remove invalid samples generated by lagging and rolling operations,
(4) Apply chronological data partitioning,
(5) Fit preprocessing steps using training data only,
(6) Train the forecasting model, and
(7) Evaluate performance on unseen future observations.

```
1. Construct base features using observations for current time (t)
   (i.e. Core-S, Core-M or Core-L set variables)
2. Generate autoregressive feature of the load:
   y(t)= dif_cons_real(t)
3. Generate rolling features using past/current values only:
   roll3_z(t)= mean (z(t), z(t-1), z(t-2))
   where z is dif_cons_real(t) or any base variable
   (i.e. tmed(t), V4(t), V12(t)...)
4. Define prediction target:
   y_target = dif_cons_real(t+1)
   (using one-step forward shifting)
5. Remove NaN rows generated by lag/rolling operations
   and target shifting
6. Apply chronological train-test split
7. For each split/fold:
      a. Fit scaler using training data only
      b. Apply scaler to test data without refitting
      c. Train forecasting model
      d. Predict y_target on unseen future samples
8. Compute RMSE, MAE, and R² metrics
```

Algorithm 1. Leakage-free temporal forecasting pipeline

This design preserves strict temporal causality and confirms that the reported forecasting improvements originate from the temporal feature-engineering strategy rather than unintended information leakage.

Model implementation and hyperparameters

All forecasting experiments were conducted using the feature configurations described in Sects. "Exploratory Data Analysis (EDA)" and "conducted using the feature ", including baseline variables, autoregressive features, and rolling statistical features. The forecasting models were implemented in Python using widely adopted machine learning libraries, including XGBoost for gradient boosting and TensorFlow/Keras for deep learning models.

To ensure a fair comparison between models, consistent preprocessing and training procedures were applied across all experiments. In particular, all models were trained using the same temporally enriched feature sets, and no model was provided with additional temporal context beyond the engineered features. The hyperparameters used for the forecasting models are summarized in Table 4.

For the XGBoost model, the number of trees, tree depth, learning rate, and sampling ratios were selected to balance model complexity and generalization performance. For the recurrent neural networks, both LSTM and GRU architectures consist of a single recurrent layer with 64 hidden units, followed by a dropout layer with a rate of 0.2 to reduce overfitting. The networks were trained for 50 epochs with a batch size of 32. For LSTM and GRU models, the input tensor is constructed with a sequence length of one, resulting in an input shape of (samples, 1, features). Each time step contains the engineered temporal features, including autoregressive and rolling variables. Therefore, these models do not process long temporal sequences, but instead operate on temporally enriched feature vectors. Under this formulation, all models receive identical temporally enriched inputs, allowing the comparison to focus on differences in model architecture rather than differences in temporal input representation. This ensures a fair and controlled evaluation across model classes. This approach is consistent with forecasting studies highlighting that engineered temporal representations based on lagged and

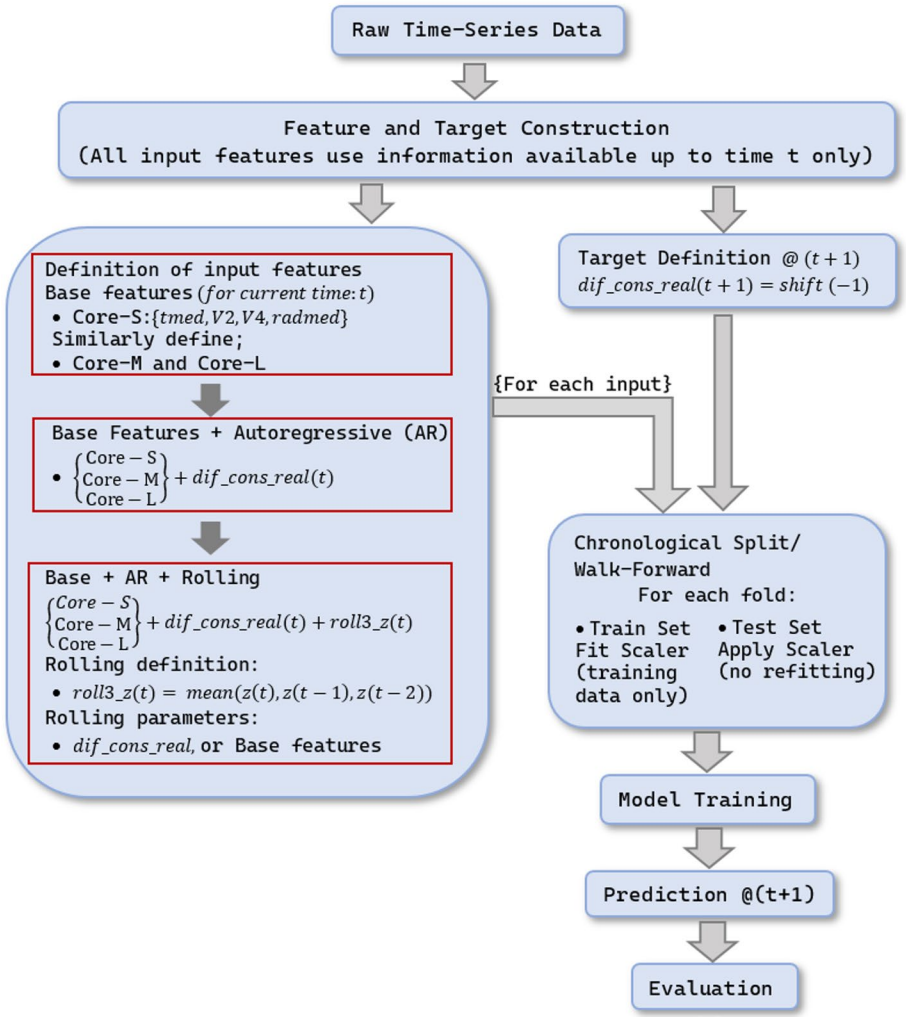


Fig. 4 Leakage-free temporal feature engineering and evaluation pipeline. All input features are constructed using information up to time t, while the prediction target corresponds to $y_{(t+1)}$. Scaling is performed using training data only, ensuring strict causality and preventing information leakage

aggregated variables can provide competitive predictive performance without necessarily relying on long sequential input windows [26, 41].

A fully connected layer with 32 neurons and ReLU activation is used before the final regression output layer. The hyperparameters were determined through preliminary exploratory experiments together with commonly adopted configurations reported in the literature. Once stable model settings were identified, the same hyperparameter configurations were consistently applied across all feature sets and validation experiments to ensure a controlled and reproducible comparison between models. In addition, fixed random seeds (RANDOM_STATE = 42) were used throughout the experiments to improve reproducibility and ensure consistent model initialization and evaluation procedures.

Performance evaluation metrics

To evaluate the predictive performance of the developed models, three widely used regression metrics were employed: Root Mean Square Error ($RMSE$), Mean Absolute Error (MAE), and the coefficient of determination (R^2). These metrics quantify the

Table 4 Hyperparameter settings used for the forecasting models

Model	Hyperparameter	Value
XGBoost	Number of trees (n_estimators)	300
	Maximum tree depth (max_depth)	5
	Learning rate	0.05
	Subsample ratio	0.9
	Column sampling (colsample_bytree)	0.9
	Objective function	reg: squarederror
LSTM / GRU	Recurrent units	64
	Dropout rate	0.2
	Dense layer units	32
	Activation function	ReLU
	Optimizer	Adam
	Loss function	MSE
	Epochs	50
	Batch size	32
	Output layer	1 neuron

differences between predicted and observed HVAC load values from complementary perspectives.

The RMSE measures the square root of the mean squared differences between predicted and observed values and is defined as

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2},$$

where y_i represents the observed value, $\widehat{y}_i$ denotes the predicted value, and n is the number of samples. RMSE penalizes large prediction errors more strongly due to the squared error term.

The MAE measures the average magnitude of the prediction errors and is defined as

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \widehat{y}_i|\,.$$

Unlike RMSE, MAE provides a more robust measure of the average prediction deviation without emphasizing large outliers.

The R^2 evaluates the proportion of variance in the observed data explained by the model and is defined as

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2}{\sum_{i=1}^{n}\left(y_i - \bar{y}\right)^2},$$

where $\bar{y}$ denotes the mean of the observed values. Higher R^2 values indicate better agreement between predicted and observed values. Together, these metrics provide complementary information regarding prediction accuracy and enable a comprehensive comparison of the forecasting models.

Results and discussion

To provide a structured analysis, the results are presented in three parts. First, the impact of temporal feature engineering on forecasting performance is examined under different feature configurations. Next, the robustness of the proposed approach is evaluated using a walk-forward validation framework. Finally, an interpretability analysis is conducted to assess the relative importance of input features.

Impact of temporal feature engineering

This section analyzes the impact of temporal feature engineering on HVAC load forecasting performance using the compact (Core-S), balanced (Core-M), and extended (Core-L) feature sets, together with autoregressive and rolling features defined in Sect. "Feature Selection and Engineering".

To clearly describe the feature composition used in the experiments, Table 5 summarizes the variables included in each temporal feature configuration. The feature configurations were constructed in a progressive, step-by-step manner to isolate the individual contribution of each feature engineering component. All included variables for each configuration use information up to the current time t.

The Baseline configuration includes only the contemporaneous variables of each core set. The Autoregressive configuration extends the baseline by incorporating the current HVAC load, dif_cons_real, as an additional predictor to capture short-term temporal dependence. The Rolling configuration further augments the feature set by including rolling mean features computed over the most recent three observations of the HVAC load, as explained in Sect. "Rolling Statistical Features", enabling the model to capture short-term consumption trends.

Finally, the Extended configuration generalizes this approach by computing rolling mean features not only for the HVAC load but also for core environmental and operational variables, providing a richer representation of short-term system dynamics. This progressive design allows the contribution of each temporal feature engineering step to be evaluated in a controlled and interpretable manner across different feature set complexities.

Table 5 Feature composition summary for the baseline, autoregressive, rolling-load, and extended rolling configurations. The configurations were constructed progressively to isolate the contribution of each feature engineering component. All included variables use information up to the current time t

Configuration	Description	Included Variables
Baseline	Contemporaneous variables only	Core-S: $\{tmed,\ V4,\ V12,\ radmed\}$ Core-M: $\{tmed,\ V4,\ V12,\ V2, radmed,\ dewpt,\ Season_3\}$ Core-L: $\{tmed,\ V4,\ V12,\ V2,\ V26,\ radmed,\ dewpt, Season_3,\ dpv\}$
Autoregressive (AR)	Baseline + current load	Baseline + $\{dif_cons_real\}$
Rolling (Load only)	Autoregressive + rolling load	AR + $\{roll3_dif_cons_real\}$
Extended (Full Rolling)	Rolling applied to all variables	AR + $\{roll3_dif_cons_real\}$+ rolling of all Core variables (e.g., $roll3_tmed, roll3_V2, \ldots$)

Table 6 presents the forecasting performance of the models under different temporal feature configurations for the compact (Core-S), balanced (Core-M), and extended (Core-L) feature sets.

Under the Baseline configuration, where no historical load information is included, all models exhibit limited predictive performance. The maximum R^2 value achieved is approximately 0.34 using the XGBoost model with the Core-S feature set. While this result suggests that even a compact feature set can provide a reasonable approximation of HVAC load behavior, the overall low accuracy across all models clearly indicates that a forecasting scenario without temporal information is insufficient for capturing the dynamics of HVAC energy consumption.

A substantial improvement is observed in the Autoregressive configuration, where the most recent load value $dif_cons_real\,(t)$ is introduced as an additional predictor. In this case, the R^2 values increase significantly across all feature sets, exceeding 0.70 for most configurations. The best performance is achieved by the XGBoost model with the Core-M feature set ($R^2 \approx 0.76$), closely followed by the Core-L configuration ($R^2 \approx 0.76$). This result highlights that the primary performance gain is driven by the inclusion of temporal information rather than by increasing feature set size.

These findings also indicate that including all available environmental and meteorological variables does not necessarily lead to better performance. Instead, selecting representative variables from highly correlated groups provides a more effective feature design, consistent with the correlation analysis in Sect. "Exploratory Data Analysis (EDA)".

In the Rolling (Load only) and Extended configurations, where rolling mean features are introduced, the performance changes are relatively modest compared to the autoregressive case. While some models exhibit slight improvements and others show minor decreases, the overall impact of rolling features remains limited. The best performance in these configurations is achieved with R^2 values around 0.79, obtained by the XGBoost and GRU models under both Core-L and Core-M feature sets.

Overall, these results reveal three important observations. First, the inclusion of recent load history is the dominant factor in improving forecasting accuracy. Second, expanding the feature set beyond a balanced configuration yields only marginal gains, indicating diminishing returns due to feature redundancy. Third, once temporal dependencies are properly captured, different model architectures achieve comparable performance, suggesting that feature engineering plays a more critical role than model selection in short-term HVAC load forecasting.

While the above results demonstrate the effectiveness of temporal feature engineering under a single train–test split, it is important to assess whether these findings generalize across different temporal segments of the dataset. In real-world building energy applications, seasonal variations and operational changes may influence model performance over time.

To address this, a robustness analysis was conducted using an expanding-window walk-forward validation framework, as described in Sect. "Experimental Setup". This approach enables performance evaluation across multiple temporal folds, reducing dependence on a single train–test boundary and providing a more reliable assessment of model generalization.

Table 6 Forecasting performance under different feature configurations.

Configuration	Core Set	Model	Number of Features	RMSE	MAE	R^2
Baseline	Core-S	XGBoost	4	1.435	1.157	0.335
		LSTM	4	1.628	1.416	0.146
		GRU	4	1.759	1.560	0.002
	Core-M	XGBoost	7	1.674	1.329	0.096
		LSTM	7	1.836	1.612	-0.087
		GRU	7	1.867	1.622	-0.125
	Core-L	XGBoost	9	1.658	1.322	0.114
		LSTM	9	1.661	1.435	0.111
		GRU	9	1.769	1.529	-0.009
Autoregressive	Core-S	XGBoost	5	0.955	0.736	0.706
		LSTM	5	1.134	0.971	0.586
		GRU	5	0.970	0.763	0.697
	Core-M	XGBoost	8	0.863	0.669	0.760
		LSTM	8	0.909	0.715	0.733
		GRU	8	0.917	0.730	0.729
	Core-L	XGBoost	10	0.867	0.673	0.757
		LSTM	10	0.920	0.736	0.727
		GRU	10	0.909	0.722	0.734
Rolling (Load only)	Core-S	XGBoost	6	0.877	0.654	0.752
		LSTM	6	0.900	0.695	0.739
		GRU	6	0.896	0.682	0.741
	Core-M	XGBoost	9	0.813	0.617	0.787
		LSTM	9	0.841	0.632	0.772
		GRU	9	0.859	0.662	0.762
	Core-L	XGBoost	11	0.790	0.593	**0.799**
		LSTM	11	0.827	0.619	0.779
		GRU	11	0.839	0.628	0.773
Extended (Full Rolling)	Core-S	XGBoost	10	0.823	0.619	0.781
		LSTM	10	0.834	0.618	0.776
		GRU	10	0.834	0.630	0.776
	Core-M	XGBoost	16	0.790	0.592	**0.799**
		LSTM	16	0.814	0.607	0.786
		GRU	16	0.817	0.610	0.784
	Core-L	XGBoost	20	0.802	0.608	0.793
		LSTM	20	0.800	0.589	0.794
		GRU	20	0.789	0.576	**0.799**

Bold values indicate the best overall predictive performance among the compared feature configurations and models

Robustness analysis via walk-forward validation

To further evaluate the robustness of the proposed feature engineering strategy, a walk-forward validation approach was employed using the best-performing temporal configurations identified in the main experiments. Robustness analysis was assessed not only across temporal folds but also across different feature set complexities. Specifically, the Core-S, the Core-M, and the Core-L baseline feature sets with Extended (Full Rolling) configurations were evaluated under identical walk-forward validation settings.

In the walk-forward validation setup, an expanding-window strategy was used. The first 50% of the chronologically ordered samples was used as the initial training set, and the following 10% was used as the first test segment. The training window was then expanded by 10% at each fold, while the next 10% segment was used for testing. This resulted in five sequential evaluation folds: 50–60%, 60–70%, 70–80%, 80–90%, and

90–100% test periods. Thus, each model was evaluated on multiple future time segments while preserving the temporal order of the data.

For each core feature set, the same model configurations and hyperparameters were used as in the main experiments, ensuring that differences in performance reflect only the impact of data partitioning and feature composition.

Table 7 presents the mean and standard deviation of RMSE, MAE, and R^2 across all folds. Reporting both average performance and dispersion provides a comprehensive assessment of model stability under varying temporal conditions and feature complexities. The results show that performance remains relatively stable across different feature set complexities. Increasing the number of features from 10 in Core-S to 16 in Core-M and 20 in Core-L results in only minor changes in performance. For example, the mean R^2 of XGBoost increases from 0.682 for Core-S to 0.692 for Core-M and 0.694 for Core-L. This indicates that feature expansion provides only marginal improvements under realistic temporal validation. Therefore, the results support the presence of diminishing returns with increasing model input complexity.

It is important to note that the performance obtained under walk-forward validation is lower than that reported single train–test split in Table 6. This difference is expected, as single-split evaluation may lead to optimistic estimates when training and testing data share similar temporal characteristics. In contrast, walk-forward validation introduces more realistic and challenging forecasting scenarios by evaluating the model across multiple temporal segments with varying data distributions and training sizes. Therefore, the results presented in Table 7 provide a more reliable assessment of the model's generalization capability.

These findings suggest that a balanced feature configuration such as Core-M may provide a practical trade-off between predictive accuracy and model complexity, while the extended Core-L configuration yields only marginal additional gains. Overall, the results indicate that the proposed forecasting framework maintains relatively stable performance across different temporal partitions and feature set complexities. Furthermore, the relatively small performance differences observed across model architectures—particularly the stable behavior of XGBoost across multiple configurations—suggest that temporally enriched feature representations play an important role in short-term HVAC load forecasting under the evaluated experimental setup.

Table 7 Walk-forward validation results for the Core-S, the Core-M, and the Core-L baseline sets with Extended (Full Rolling) configurations. Results are reported as mean ± standard deviation across folds

Core Set	Model	Number of Features	RMSE (mean ± std)	MAE (mean ± std)	R^2 (mean ± std)
Core-S	XGBoost	10	1.195 ± 0.481	0.860 ± 0.311	0.682 ± 0.046
Core-S	LSTM	10	1.172 ± 0.393	0.846 ± 0.269	0.686 ± 0.022
Core-S	GRU	10	1.193 ± 0.406	0.868 ± 0.268	0.675 ± 0.020
Core-M	XGBoost	16	1.170 ± 0.464	0.838 ± 0.280	0.692 ± 0.054
Core-M	LSTM	16	1.200 ± 0.389	0.888 ± 0.239	0.666 ± 0.041
Core-M	GRU	16	1.212 ± 0.462	0.901 ± 0.305	0.666 ± 0.050
Core-L	XGBoost	20	1.169 ± 0.468	0.843 ± 0.283	0.694 ± 0.056
Core-L	LSTM	20	1.214 ± 0.484	0.905 ± 0.331	0.667 ± 0.056
Core-L	GRU	20	1.173 ± 0.374	0.865 ± 0.217	0.679 ± 0.047

Table 8 Effect of lag length on forecasting performance (Core-M with Extended (Full Rolling), XGBoost)

Lag Configuration	Number of Features	RMSE	MAE	R^2
$y_{(t)}$	16	0.790	0.592	0.799
$y_{(t)}$, $y_{(t-1)}$	17	0.779	0.583	0.804
$y_{(t)}$, $y_{(t-1)}$, $y_{(t-2)}$	18	0.787	0.579	0.800

Table 9 Effect of rolling window size on forecasting performance (Core-M with Extended (Full Rolling), XGBoost)

Window Size	RMSE	MAE	R^2
3	0.790	0.592	0.799
6	0.870	0.685	0.756
12	0.974	0.787	0.694

Extended ablation analysis

To further validate the robustness of the proposed feature engineering strategy, an extended ablation study was conducted using one of the best-performing configurations identified in the main experiments, namely the Core-M feature set with Extended (full rolling) temporal features evaluated using the XGBoost model. The reference configuration achieves an R^2 of approximately 0.799, and none of the tested variations produce substantial improvements beyond this level, indicating that the proposed temporal representation is already close to a stable optimum.

First, the effect of autoregressive lag length was examined by progressively increasing the number of past load observations included in the model. As shown in Table 8, extending the lag structure from only the current load $y_{(t)}$ to include one additional lag $y_{(t-1)}$ results in a modest performance improvement ($R^2 \approx 0.804$). However, further increasing the lag depth to include $y_{(t-2)}$ does not provide additional benefit and slightly reduces performance. This behavior suggests that the effective temporal memory of the HVAC system is relatively short, and that recent load history already captures most of the predictive temporal dynamics.

Next, the impact of rolling window size was analyzed. The results, presented in Table 9, show a clear degradation in performance as the rolling window length increases. While a short window of size 3 preserves short-term temporal variations and achieves the best overall balance, larger windows (6 and 12) introduce excessive smoothing and significantly reduce forecasting accuracy. This confirms that short-term HVAC dynamics are critical for accurate prediction and should not be excessively averaged over long temporal horizons.

The influence of rolling-statistics design was further investigated in Table 10. Extending the rolling features from mean-only statistics to include rolling standard deviation improves the forecasting performance and produces the best observed result ($R^2 \approx 0.805$). This indicates that short-term variability information provides complementary predictive value in addition to average temporal trends. However, further extending the rolling representation by including minimum and maximum statistics does not improve performance and slightly reduces accuracy. This observation suggests that increasing feature complexity does not automatically lead to better generalization, and that the combination of rolling mean and standard deviation already captures the most relevant short-term temporal behavior.

Table 10 Effect of rolling statistics on forecasting performance (Core-M with Extended (Full Rolling), XGBoost; window = 3)

Rolling Statistics	Number of Features	RMSE	MAE	R^2
Mean only	16	0.790	0.592	0.799
Mean + Std	24	0.778	0.581	0.805
Mean + Std + Min + Max	40	0.784	0.590	0.802

Table 11 Effect of seasonal indicator on forecasting performance (Core-M with Extended (Full Rolling), XGBoost)

Configuration	Number of Features	RMSE	MAE	R^2
With Season_3	16	0.790	0.592	0.799
Without Season_3	14	0.802	0.608	0.792

Table 12 Effect of occupancy-related operational proxy variables on forecasting performance (Core-M with Extended (Full Rolling), XGBoost)

Configuration	Number of Features	RMSE	MAE	R^2
Full Core-M configuration	16	0.790	0.592	0.799
Without V4	14	0.787	0.570	0.800
Without V12	14	0.828	0.614	0.779
Without V4 and V12	12	0.815	0.578	0.786

The contribution of calendar-related information was evaluated by removing the seasonal indicator variable (Season_3). As shown in Table 11, excluding this variable leads to a small but noticeable performance decrease, indicating that seasonal context provides useful auxiliary information. Nevertheless, the relatively limited performance drop confirms that temporal load-history features remain the dominant source of predictive capability.

Since the PLEIA dataset does not contain direct occupancy measurements, HVAC operational variables such as the number of active HVAC devices (V4) and HVAC control/setpoint variables (V12) were considered as indirect occupancy-related proxies. Their contribution was analyzed through an additional operational-proxy ablation study summarized in Table 12. Interestingly, removing V4 does not significantly degrade the performance and even produces a marginal improvement, suggesting that its information content is largely captured by temporal load-history and rolling features. In contrast, removing V12 causes a more noticeable performance reduction, indicating that HVAC control behavior contains useful contextual information. Removing both variables simultaneously further decreases the model accuracy, although the overall degradation remains moderate compared to the impact of temporal-feature design choices.

Overall, the extended ablation analysis demonstrates that while minor performance gains can be achieved through careful tuning of temporal feature design, the overall model behavior remains highly stable across different configurations. The results consistently confirm that the primary driver of predictive accuracy is the inclusion of recent temporal information, rather than deeper lag structures, larger rolling windows, or excessive feature complexity.

Interpretability analysis

To interpret the contribution of individual variables, permutation feature importance analysis was performed using the XGBoost model under the best-performing feature

configuration (Core-M with Extended (Full Rolling)). As illustrated in Fig. 5, the permutation importance analysis confirms that the current HVAC load y(t) is by far the most influential predictor. Although the rolling load feature remains important, the Core-M Extended configuration also highlights the relevance of solar radiation and its rolling representation. This suggests that, once recent load history is included, short-term solar gains provide complementary information for one-step-ahead HVAC load prediction. Nevertheless, the contribution of.

$y_{(t)}$ remains substantially larger than that of all other predictors, supporting the conclusion that recent temporal load information is the dominant driver of forecasting performance.

To further examine the predictive behavior of the best-performing model, additional diagnostic visualizations were analyzed. Figure 6 presents the comparison between the measured HVAC load and the predicted values. The predicted series closely follows the temporal variation of the observed load and successfully captures the dominant short-term consumption dynamics. Small deviations are mainly observed around abrupt load transitions and peak-consumption periods, which are inherently more difficult to predict in short-term HVAC forecasting tasks.

Furthermore, Fig. 7 shows the distribution of the residual errors. The residuals are approximately centered around zero, indicating the absence of strong systematic bias in the forecasting results. Although the residual distribution exhibits slight asymmetry and a moderate positive tail, the overall error distribution remains reasonably concentrated around the zero-error region. This behavior suggests that the proposed framework provides stable short-term forecasting performance while still exhibiting occasional larger deviations during rapidly changing operating conditions.

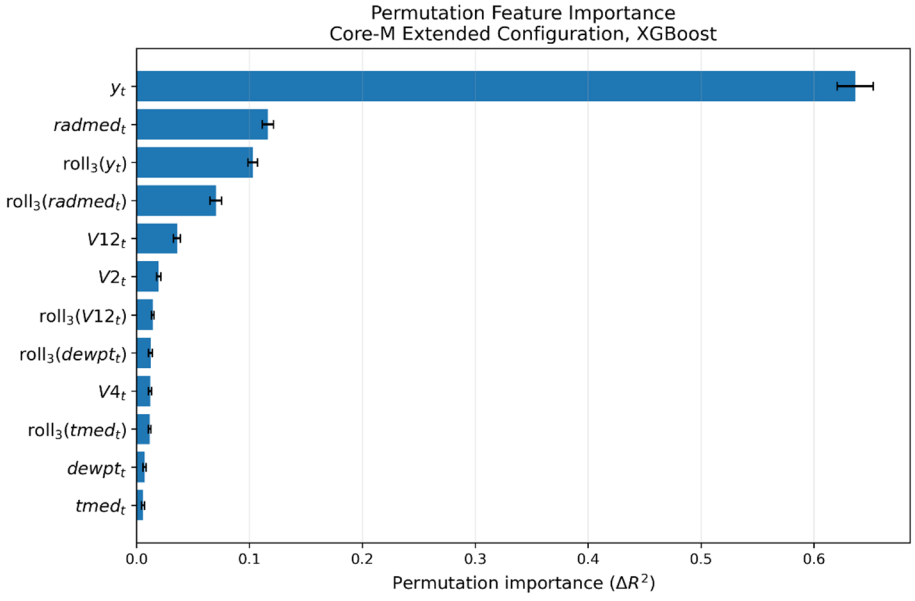


Fig. 5 Permutation feature importance (ΔR2) obtained from the XGBoost model under the Core-M Extended configuration. The current HVAC load $y_{(t)}$ is the dominant predictor, confirming the strong short-term temporal dependence of HVAC energy consumption. The rolling load feature $roll3_y_{(t)}$, solar radiation $radmed$, and its rolling representation $roll3_radmed_{(t)}$ provide secondary contributions, while indoor, control, humidity, and operational variables have comparatively smaller effects

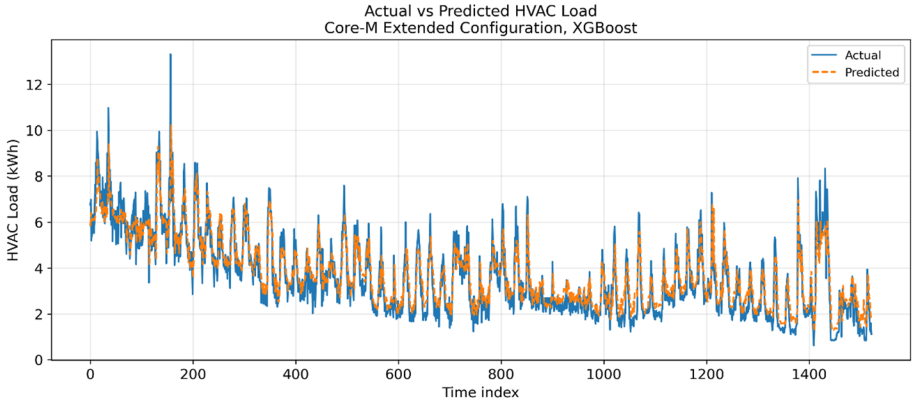


Fig. 6 Comparison between the measured HVAC load and the predicted values obtained from the best-performing forecasting model. The predicted series closely follows the temporal variation of the measured load, indicating that the model successfully captures short-term load dynamics

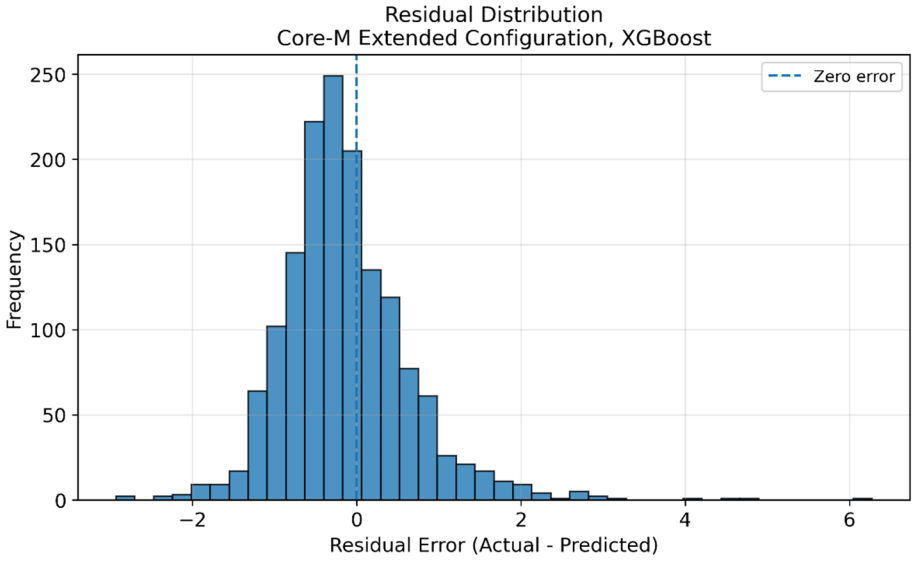


Fig. 7 Distribution of residual errors for the best-performing model. The residuals are approximately centered around zero, indicating that the model errors do not exhibit significant systematic bias

Model limitations and future research directions

Despite the encouraging forecasting results obtained in this study, several limitations should be acknowledged.

First, the experiments were conducted using only Building A from the PLEIAData complex. Although this controlled setup enabled a systematic investigation of temporal feature-engineering strategies, the transferability of the proposed framework across different buildings, occupancy patterns, HVAC control policies, and climatic conditions was not explicitly evaluated. Therefore, the reported results should be interpreted primarily as evidence of the effectiveness of temporal feature representations within the studied operational context rather than as a universally validated building-energy forecasting framework.

Second, the present study focuses exclusively on one-step-ahead forecasting. While short-horizon prediction is highly relevant for real-time HVAC control, rolling optimization, and demand-response applications, longer forecasting horizons may exhibit

different temporal dependencies and uncertainty characteristics. Consequently, the effectiveness of the proposed temporal feature-engineering strategy under multi-step or long-horizon forecasting scenarios remains an open research question.

Another limitation concerns the representation of sequential learning in the recurrent models. In this work, LSTM and GRU models operate using a sequence length of 1, where each sample consists of temporally engineered tabular features rather than long sequential input windows. As a result, the recurrent architectures primarily function as nonlinear regressors over engineered temporal representations instead of fully exploiting long-term sequence-learning capabilities.

In addition, the PLEIA dataset does not contain direct occupancy measurements. Although several HVAC operational variables were considered as indirect occupancy-related proxies, the influence of explicit occupant behavior and occupancy-driven dynamics could not be fully analyzed.

Nevertheless, the results consistently demonstrate that recent temporal load information provides the dominant contribution to forecasting performance across multiple feature configurations, rolling-window choices, and model architectures. Moreover, even moderate forecasting improvements may still provide meaningful operational value in practical energy-management systems when applied continuously over long operational periods.

Recent forecasting studies in adjacent application domains have increasingly emphasized the importance of robustness-oriented evaluation, explainable AI, and interpretable forecasting methodologies. Hybrid deep learning architectures, explainable forecasting frameworks, and physics-informed learning approaches have been successfully explored for environmental and climate-related forecasting problems, including temperature prediction, air-quality forecasting, and emission modeling [42–46]. These studies highlight the growing importance of reliable and interpretable forecasting frameworks beyond purely accuracy-oriented performance evaluation.

Future research may therefore focus on; (i) cross-building validation, (ii) multi-horizon forecasting, (iii) transferability across different HVAC control regimes, (iv) integration of direct occupancy measurements, and (v) hybrid sequence-learning architectures capable of jointly exploiting engineered and long-horizon temporal dependencies.

In particular, incorporating longer temporal windows and multi-scale rolling representations may help capture medium- and long-term HVAC consumption dynamics that are not fully represented in one-step-ahead forecasting. Furthermore, advanced sequence-based architectures such as Transformer models may provide additional insight into complex temporal interactions under longer forecasting horizons. Finally, combining interpretable machine learning techniques with building-physics knowledge may further improve both forecasting reliability and model interpretability in practical HVAC energy-management applications.

Conclusion

This study investigated the role of temporal feature engineering in short-term HVAC load forecasting using the PLEIAData building dataset. Three forecasting models representing different learning paradigms—XGBoost, LSTM, and GRU—were evaluated under multiple progressively enriched temporal feature configurations, including autoregressive and rolling statistical representations.

The experimental results demonstrated that models relying only on contemporaneous environmental and operational variables exhibit limited predictive capability. In contrast, incorporating temporally enriched representations based on recent HVAC operational history substantially improves forecasting performance across all evaluated model classes. In particular, autoregressive load information and rolling statistical descriptors consistently enhanced predictive accuracy by capturing short-term consumption dynamics and operational continuity.

Among the evaluated configurations, the Core-M Extended feature configuration combined with XGBoost achieved one of the best and most stable overall forecasting performances, reaching an R^2 value of approximately 0.799 under the primary experimental setup. Additional walk-forward validation experiments further demonstrated that the proposed forecasting framework maintains relatively stable behavior under varying temporal partitions and feature set complexities. Although increasing feature complexity provides moderate improvements, the results also indicate diminishing returns beyond a balanced temporal representation.

Interpretability analysis based on permutation feature importance revealed that the current HVAC load $y_{(t)}$ is the dominant contributor to forecasting performance, while environmental variables and rolling statistical features provide complementary information for capturing short-term operational and thermal variability.

Overall, the findings suggest that temporal feature representation plays a central role in short-term HVAC load forecasting and can strongly influence predictive performance independently of the underlying model architecture. The study further highlights the importance of leakage-aware evaluation, robustness-oriented validation, and interpretable forecasting analysis for developing reliable building energy prediction frameworks.

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Author contributions
S.E. conceived the study, performed the experiments, analyzed the results, and wrote the manuscript.

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Data availability
The dataset analyzed during the current study (PLEIAData building dataset) is publicly available in the Zenodo repository under record number 7620136 and can be accessed at: https://doi.org/10.5281/zenodo.7620136.

Declarations

Ethics approval and consent to participate
Not applicable. The study does not involve human participants or animals.

Competing interests
The authors declare no competing interests.

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